MILITARY SPECIFICATION

EXCAVATOR, MULTIPURPOSE, WHEEL-MOUNTED: FOUR-WHEEL DRIVE, UNIVERSAL BOOM

This specification is approved for use the Marine Corps, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a multipurpose, wheel-mounted, diesel powered, pneumatic tired, mobile, four-wheel drive, hydraulic excavator, and optional front end equipment for excavating, trenching, ditching, sloping, grading, backfilling, rip rap work and loading with mobile capabilities over rough terrain and highways.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

F-P-666	- Primer, Pressure, Internal Combustion Engine (Cold Starting Aid) and Primer Cartridges
VV-F-800	- Fuel Oil, Diesel

MILITARY

MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3805

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MIL-P-514	- Plate, Identification, Instruction and Marking, Blank
MIL-T-704	- Treatment and Painting of Materiel
MIL-L-2104	- Lubricating Oil, Internal-Combustion Engine, Heavy Duty
MIL-L-2105	- Lubricating Oil, Gear, Multi-Purpose
MIL-C-3580	- Cranes, and Crane Shovels, Truck, Crawler, & Wheel Mounted, Full Revolving and Their Attachments, Packaging of
MIL-T-5624	- Turbine Fuel, Aviation, Grade JP-4 and JP-5
MIL-G-10924	- Grease, Automotive and Artillery
MIL-M-43719	- Marking Materials, and Markers, Adhesive, Elastomeric, Pigmented, General Specification
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic
MIL-M-46278	- Mirror Assembly, Rearview: Automotive, Exterior Mounting
MIL-G-46795	- Generator System, Alternator, Rectifier, 60 Amperes, 28 Volts
MIL-E-52798	- Enamel, Alkyd, Camouflage
MIL-C-62122	- Cable Assembly, Slave, Electrical, Connectors, Plug and Receptacle

STANDARDS

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics-Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement
MS-35000	- Battery, Storage, Lead-Acid, Waterproof
MS-35746	- Coupling, Automotive, Air Brake Lines
MS-51118	- Pintle Assembly, Towing - 40,000 Lbs Capacity, Manual Release

DRAWINGS

Ordnance Corps (Army Tank Automotive Command)
8712407 Rifle Mounting Kit

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC (NMFTA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, 1616 P Street, N.W., Washington, D.C. 20036.)

THE TIRE AND RIM ASSOCIATION, INC.

Tire & Rim Year Book

(Application for copies should be addressed to the Tire & Rim Association Incorporated, 3200 W. Market Street, Akron, Ohio 44313.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J296	Hoe Dipper (Backhoe Bucket) Rating
J429	Mechanical and Material Requirements for External Threaded Fasteners
J525	Welded and Cold Drawn Low Carbon Steel Annealed for Bending and Flaring
J695	Turning Ability and Off Tracking - Motor Vehicles
J742	Front End Loader Bucket Rating
J816	Engine Test Code
J1097	Hydraulic Excavator Lift Capacity Ratings
J1152	Minimum Performance Criteria for Braking Systems for Off-Highway, Rubber-Tired, Self-Propelled Construction Machines
J1166	Operator Station Sound Level Measurement Procedure for Powered Mobile Earthmoving Machinery

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)

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POWER CRANE SHOVEL ASSOCIATION (PCSA)

Standard No. 3 - Mobile Hydraulic Excavator

(Application for copies should be addressed to the Power Crane Shovel Association, Marine Plaza, Suite 1700, 111 E. Wisconsin Avenue, Milwaukee, WI 53202.)

Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

3. REQUIREMENTS

3.1 First article. One complete excavator and the front end equipment ordered, conforming to all requirements of this specification, shall be submitted for first article inspection and tests. Approval of the first article shall not relieve the supplier of the responsibility of furnishing equipment in accordance with this specification. The first article shall remain at the manufacturer's plant as a sample and shall be delivered as the last unit of the initial contract.

3.2 Material. Subject to the provisions of this specification, and except as specifically specified herein, all materials, components, and design shall be identical to that currently sold by the manufacturer. Rubber products furnished, including tires, shall not exceed eight calendar quarters in age from cure date to date of acceptance of each excavator by the Government.

3.2.1 Fungus and moisture resistance. All electrical circuitry connections and components of all electrical systems, except those noted below shall be coated with varnish conforming to MIL-V-173:

(a) Components and elements inherently inert to fungi or hermetically sealed.

(b) Components and elements whose operation would be adversely affected by the application of varnish.

3.3 Design. The excavator shall be a diesel-powered, self-propelled, full-revolving, four-wheel drive machine with oscillating front axle, four stabilizers or outriggers, universal boom with an extendable three-position tool boom and tool angling device for operation with optional front end equipment.

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3.3.1 Characteristics. The excavators shall conform to the following characteristics:

Weight, minimum (pounds(lbs))[1/]	32,000
Weight, maximum (lbs)[1/]	43,000
Transport Height, maximum (inch(in))	160
Transport Width, maximum (in)	96
Side Tail Swing, maximum (in)	80
Transport Length, maximum Boom Extended (in)	470
Transport Length, maximum Boom Retracted (in)	420
Ground Clearance, minimum (in) not less than	14
Transport Vehicle Clearance Circle Diameter, Tool Boom Retracted maximum (feet(ft))[2/]	80
Angles of Approach and Departure, minimum (degrees)	25

3.3.2 Minimum requirements. Minimum excavator requirements shall be as follows:

Digging Force (lbs)	
(a) Tool boom arm force (retracted), minimum	9,200
(b) Bucket tangential force, minimum	10,500
Drawbar pull (lbs), lowest gear range, minimum	14,000
Swing stall torque, (ft/lb), minimum	16,800
Travel speed, miles per hour (mi/h)	
low range, minimum	0 - 4
high range, minimum	0 -22

[1/] Weight shall include fully serviced and fueled excavator with excavator bucket, pintle hitch, tires as specified and counterweight furnished as an integral part of the excavator, but shall not include operator.

[2/] Vehicle clearance circle in accordance with SAE J695.

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Cycle Times, No load (seconds), minimum

Hoist extend	6
Hoist retract	7
Crowd extend	9
Crowd retract	7
Tool extend	5
Tool retract	4
Stabilizers extend	6
Stabilizers retract	6
360 degree swing	14

3.3.3 Lift capacities. Lift capacities specified without tool and tilt machanism shall not exceed the machine's structural competence, nor 75 percent of the stability or 87 percent of the hydraulic capacity of the machine in compliance with SAE J1097 for 360 degree rotation, at the load point height, radius and tool boom position shown in Table I. The capacity to lift a minimum of 10,000 lbs to a 15-ft height at a 15-ft minimum radius shall enable the machine to life preformed bunkers and 8-ft x 8-ft x 20-ft shelters from trailers and place them into an excavation. The machine must have the ability to perform this task with the excavating bucket attached. To accommodate the load-lifting slings, the manufacturer may install an additional lifting point on the working stick or boom which does not have to meet the requirements of SAE J1097 for this 10,000-lb lift.

TABLE I - LIFT CAPACITY

LOAD POINT HEIGHT, FEET	RADIUS 10 FEET	15 FEET	20 FEET	25 FEET
	TOOL BOOM EXTENDED			
+20				
+15		(See 3.3.3)		4,620
+10				4,680
+ 5			5,920	4,580
Ground		8,960	6,340	4,500
- 5	16,000	9,650	6,210	
-10	15,200	9,550		

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LOAD POINT HEIGHT, FEET	RADIUS 10 FEET	15 FEET	20 FEET	25 FEET
		TOOL BOOM RETRACTED		
+20				
+15		(See 3.3.3)	5,700	
+10			6,240	
+ 5		8,740	6,360	
Ground	15,700	9,700	6,230	
- 5	15,600	9,590		

3.3.4 Digging requirements. The excavator, when equipped with the 5/8-cubic-yard excavating bucket, less tool angling device, shall meet digging requirements as follows:

Tool Boom Fully Extended (ft)	
Maximum reach at ground level (not less than)	28'6"
Minimum reach at ground level (not more than)	10'0"
Maximum digging depth (not less than)	17'10"
Maximum digging depth, 8 ft level 1/ (not less than)	17'5"
Maximum dump clearance 2/ (not less than)	19'0"
Maximum bucket teeth height 3/ (not less than)	27'0"
Tool Boom Fully Retracted (ft)	
Maximum reach at ground level (not less than)	25'0"
Minimum reach at ground level (not more than)	7'6"
Maximum digging depth (not less than)	13'10"
Maximum digging depth, 8 ft level 1/ (not less than)	13'0"
Maximum dump clearance 2/ (not less than)	17'0"
Maximum bucket teeth height 3/ (not less than)	25'0"

1/ Maximum depth of cut for 8 ft level bottom (straight cleanup.)

2/ Maximum vertical clearance of bucket from grade with tool at max. height.

3/ Bucket teeth distance from grade at end of highest dump.

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3.3.5 Stress. The design of the excavator and front end equipment shall be such that the maximum unit stress imposed by the rated loading throughout the working area shall not exceed 70 percent of the yield point of any structural member.

3.3.6 Revolving upper structure. The upper structure shall generally consist of the universal boom, hydraulic actuators, platform, turntable, swing mechanism, and operator's cab. Rearview mirrors, conforming to type II of MIL-M-46728, shall be mounted on the right and left sides of the upper structure to provide optimum vision to the rear of the operator.

3.3.6.1 Universal boom. The boom shall be designed for universal use with excavator buckets, front loading buckets, digging buckets, and other such front end equipment. The boom shall consist of main boom actuated by hoist cylinder, a tool boom actuated by crowd cylinder, tool boom extension, tool cylinder and tool angling device. The extension shall have a minimum of three positions for a maximum extension of not less than 4 ft. The boom shall be constructed of welded steel reinforced in all areas of stress concentration. Replaceable hardened steel bushings shall be installed at pivot points subject to extensive wear. Lubrication fittings, where appropriate, shall be located for convenient servicing.

3.3.6.2 Tool angling device. A hydraulically actuated, and operater controlled, device shall be mounted on the tool boom for supporting and rotating mounted tools or buckets at any angle up to 40 degrees to the right (clockwise direction) or 40 degrees to the left (counter clockwise direction) through a total angle of 80 degrees in addition to a normal bucket sweep angle of 155 degrees, minimum. The operating control shall be integrated with other controls to simultaneously provide the operator with positive control of tool angle at all times. The design shall interface with standard boom and bucket mounts so that the tools or buckets may be mounted directly to the tool boom extension after removal of the tool angling device. The design shall be adequate to withstand all normal forces, shall provide for positive position control without creep, and shall weigh not more than 700 lbs exclusive of the hydraulic lines and control.

3.3.6.3 Platform. The upper structure revolving platform shall be of balanced design with minimum tail swing over side. The turntable mount shall be designed for efficient transfer of forces to the lower structure with a minimum of distortion.

3.3.6.4 Swing assembly. The upper structure shall revolve on an outer race bolted to an accurately machined and ground mounting face on the upper structure frame. A continuous internal gear shall be an integral part of the inner race that fully encloses and interlocks with the upper race to provide an enclosed low friction turntable bearing driven by a reversible hydraulic motor with gear reduction drive and built-in disc brake. A conveniently located lubrication fitting shall be provided in the operator's cab. The swing driver motor gear reducer

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shall run submerged in oil. The upper structure swing braking shall normally be provided by the hydraulic motor. The mechanical brake actuated by the operator shall hold the position of the upper structure while in the operating position.

3.3.6.5 Operator's cab. The operator's cab shall be mounted on the left side of the boom to provide maximum visibility and natural driving position while in the operating mode. The cab shall be all steel, fully enclosed and weatherproof, with windows on all sides. The windshield shall be safety glass with upper section storable overhead. Nonbreakable glass or safety glass will be furnished in side and rear windows. Bifold or sliding door shall be positively retained in all positions. Operator's seat shall be contoured bucket type with three-position back cushions and 6-inch fore and aft adjustment, with seat belt of a minimum 3-inch width attached securely. A conventional steering wheel and control for reversing the steering while working over rear shall be provided. The cab shall be equipped with a two-speed electrical windshield wiper and a minimum 7,000 British thermal units per hour diesel fueled cab heater with window defogger(s). When tested in accordance with SAE J1166, the excavator shall meet a sound level of 85 decibels(dB)(A). If the SAE J1166 sound level exceeds 85dB(A), the following warning plate shall be permanently affixed to the excavator in a conspicuous location in the operator's compartment "WARNING - EAR PROTECTION REQUIRED WHEN OPERATING THIS EXCAVATOR". The sound level shall not exceed 90dB(A) at the operator's ear when tested in accordance with SAE J1166.

3.3.6.6 Operator's controls and instruments. All operator controls for lower and upper works shall be located in the operator's cab. All excavator functions shall be controllable while upper structure is being driven through 360-degree continuous rotation. All controls for the operation of the upper works shall be hand operated. Controls shall include crowd control, hoist auxiliary, fast hoist, tool control, swing control, accelerator pedal, steering wheel, service brake pedal, emergency engine shut off, turn signal/emergency flasher with indicator, horn, house brake control, steering reverse lever, parking brake control, gear selector, range selector lever, forward/reverse selector, accelerator lock, digging brake control, and outrigger/stabilizer controls. Instruments in the operator's cab shall include low air pressure warning light, high engine temperature warning light, engine oil pressure, and high torque convertor oil temperature gages or warning lights, cab instrument panel light and switch, winshield wiper switch, clearance lights switch, starter button, and ignition switch. Controls and instruments required for options are in addition to the above. All in-cab instruments and controls shall be within reach, and clearly visible during day or night operations, from the operator's seat.

3.3.6.7 House lock. The excavator shall be provided with a positive mechanical house lock which will preclude rotation of the revolving upper structure when engaged. Operation of the house lock shall be accomplished from the operator's compartment.

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3.3.7 Lower structure. The lower structure or chassis shall generally consist of tires, wheels, brakes, transmission, axles, frame, steering, outriggers and fenders over all wheels. The front of the lower structure shall be referred to as the engine end or forward end of the unit during the operating mode.

3.3.7.1 Tires and wheels. The tires shall be tubeless with super ground grip, L-2 tread. The same size and type shall be used on all wheels. Tires and wheels shall be in accordance with the Tire and Rim Association standards. The minimum tire size shall be 17.5 x 25, 12 ply rated, and shall not exceed loading as recommended by the Tire and Rim Association.

3.3.7.2 Service brakes. The service brakes shall be the air powered and controlled hydraulic drum or disc type located at each wheel. The air supply shall be provided by an air compressor and reservoir of suitable size to permit a minimum of six stops with a dead engine on maximum required longitudinal slope. The reservoir shall be equipped with conveniently located drain and safety relief valve. The service brakes shall be controlled simultaneously on all four wheels by a foot controlled air valve during travel or positively set during digging by a conveniently located manual valve. A light to warn the operator that the brakes are set and a gage to show air pressure shall be furnished. Additionally, the requirements of SAE J1152 for excavators of this class shall be met.

3.3.7.3 Parking brakes. A mechanically actuated parking brake(s) effective on all wheels and independent of the service brakes shall be provided. The parking brake shall be automatically set should the control pressure fail. The parking brake shall hold the excavator on a 30-percent slope.

3.3.7.4 Transmission and torque converter. The transmission shall be torque converter drive with shuttle type speed reverser and selective sliding gears or torque converter drive with power shift. The forward and reverse speed ranges shall be a minimum of two forward and two reverse available with the excavator in motion. Drawbar pull and the lowest and highest travel speed ranges shall be in accordance with 3.3.2. The transmission and torque converter fluid shall conform to MIL-L-2104.

3.3.7.5 Drive train. The drive train shall consist of drive shafts, torque proportioning differentials and outboard planetary reduction axles. The front axle shall be of the steerable type mounted to oscillate a minimum of 6 degrees. Oscillation locks shall be provided. The rear axle shall be rigidly mounted and shall be furnished with a "no spin" feature. The turning clearance diameter shall not be greater than that specified in 3.3.1.

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3.3.7.6 Steering. The steering system shall be controlled by a steering wheel. The effort required to turn the wheel shall not exceed 5 lbs at the rim with engine running at low idle and excavator not in motion. The excavator shall be capable of being steered while traveling forward or backward with the upper structure in any position. A mechanism shall be provided to reverse the steering mode while traveling or working with the boom over the rear of the excavator.

3.3.7.7 Stabilizers. Four hydraulic stabilizers/outriggers shall be furnished to provide machine stability under lift capacity conditions specified in 3.3.2. Stabilizers shall be structurally sized, positioned and mounted to stabilize the machine under all working conditions. Cycle time shall not exceed that specified in 3.3.2. Each stabilizer pad shall have an effective area of not less than 215 square inches. Mechanical locks shall be provided to prevent movement when the stabilizers are in the raised, stowed position. Hydraulic check valves shall be installed to prevent movement while the stabilizers are used when operating the excavator.

3.3.7.8 Ground clearance. The ground clearance and angles of approach and departure shall be as specified in 3.3.1.

3.3.8 Engine. An engine shall be furnished to simultaneously power the drive train and hydraulic system. The engine shall be mounted on rubber isolators to reduce noise and vibration. The engine shall be a liquid cooled diesel engine with sufficient power and speed rating so that operation of the excavator under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer. The engine shall be capable of operating and meeting the performance specified herein on diesel fuel conforming to VV-F-800 and JP-5 turbine fuel conforming to MIL-T-5624. The engine shall be designed for use in off-highway equipment and shall have replaceable cylinder liners surrounded by coolant, rotating exhaust valves, replaceable exhaust valve inserts, and a main bearing on each side of each connecting rod. A variable governor shall be furnished to control the engine at all throttle settings.

3.3.8.1 Fuel system. The engine shall be provided with a preliminary filter, hand primer pump, transfer pump and in line two stage filter between the fuel tank and the injector pump. The fuel tank with level indicator shall be mounted in a protected position. The fuel tank size shall be 50 gallons minimum or sufficient for 10 hours operation, whichever is greater. Such operation will be recognized as a combination of moving, digging, or other normal functions resulting in maximum fuel consumption.

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3.3.8.2 Engine cooling system. The liquid cooling system, consisting of radiator, air shroud, fan-water pump and thermostat, shall be of adequate capacity to permit operation under continuous operating conditions at an ambient temperature of -25°F to +125°F. The above is exclusive of supplementary radiator shroud and fan cooling capacity necessary for hydraulic cooling should the engine cooling system also be used for that purpose.

3.3.8.3 Lubrication system. The engine shall be furnished with a lubrication system of adequate size to provide staisfactory lubrication under all conditions of operation. The system shall be protected by a screen ahead of oil pump, oil cooler and a readily replaceable full flow spin-on filter. A dipstick to show oil level shall be provided.

3.3.8.4 Air intake system. The engine air intake filtration system shall consist of a dry type filter with built in or separate precleaner and recleanable element. The capacity of the air cleaner shall be in accordance with the engine manufacturer's recommendation. A visual warning device showing when service should be performed is required.

3.3.8.5 Exhaust. The engine exhaust shall be under hood, or otherwise protected, to preclude water entering the engine during high wind rain storms. A rain cap or hat is not considered adequate under these conditions.

3.3.8.6 Starter and alternator. The engine starter and alternator shall be 24 volts and of adequate size for the low temperature starting and operating conditions specified herein. The excavator shall not start until the transmission shift lever is in neutral.

3.3.8.7 Starting aid. The engine shall be provided with a measured shot ether cold weather starting aid conforming to F-P-666, type I, size 3.

3.3.8.8 Instruments and controls. The following instruments and controls shall be located on the engine shroud and shall be visible to the operator in the cab in the driving mode:

Engine oil pressure and water temperature gages
Ammeter
Hourmeter
Tachometer
Fuel level gage
Air pressure gage
Torque-converter oil temperature gage
Panel lights
Light switch to control panel lights, head lights and tail lights

3.3.9 Hydraulic system. Engine driven hydraulic pumps shall supply the power for all excavator hydraulic functions. Hydraulic circuits shall be designed to promote simultaneous operation of all functions without interference. Main relief valve settings shall be provided to protect hydraulic components.

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3.3.9.1 Hydraulic pumps. A two- or three-section positive displacement high pressure pump shall be directly connected to the engine crankshaft, or to a power take-off on the transmission. The flow and pressure requirement from each section shall be balanced against demand, and provision shall be made to unload, operate in a series, parallel and direct flow from the sections as necessary to maintain high system efficiency.

3.3.9.2 Hydraulic fluid. The hydraulic fluid used in the system shall conform to MIL-L-2104. All components and materials shall be compatible with these oils throughout an oil operating temperature range of -25°F to +225°F.

3.3.9.3 Hydraulic relief valves. Each hydraulic circuit shall be supplied with main hydraulic relief valves set for the working pressure of that circuit. Secondary relief valves shall also be supplied where necessary to protect the circuit.

3.3.9.4 Hydraulic control valves. Manually controlled spring-centered, three-position, spool type, open center, control valves shall be located adjacent to the operator's control center. Valves will have built-in main relief, load check and secondary relief valves as required. Control linkage shall be provided with antifriction bearings. Valve and control linkage adjustments shall be readily accessible for service.

3.3.9.5 Hydraulic oil reservoir. The hydraulic reservoir assembly shall include a magnetic drain plug, atmospheric breather filter, fill cap, level indicator and 100-mesh outlet screen. Reservoir shall be of sufficient size and design to absorb volume changes within the system and provide for fluid deaeration and make-up.

3.3.9.6 Hydraulic oil cooler. A hydraulic cooler shall be furnished to dissipate any heat build-up in the hydraulic system when operating the excavator under continuous operating conditions in ambient temperatures up to 125°F. Air or liquid type cooler shall be installed in a low pressure return line. A by-pass valve and protection from surge pressures shall be provided.

3.3.9.7 Hydraulic oil filtration. Readily serviceable 100-mesh or equivalent screen filters shall be supplied in the pump inlet circuit to all pump sections. In addition, a paper filter with integral by-pass rated 33 microns nominal or better shall be furnished in the discharge or return line of one of the pump sections having the largest flow.

3.3.9.8 Hydraulic oil swivel. The hydraulic swivel shall have porting of adequate size to minimize pressure drop. The design shall also include lubrication, numbered ports and replaceable seals with back-up rings for servicing.

3.3.9.9 Hydraulic actuators. The design shall use a minimum number of hydraulic actuators or cylinders consistent with force, stability, cycle

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time and visibility requirements. The cylinders shall be of heavy duty construction and shall have replaceable mounting bushings and surface hardened and chrome plated rods.

3.3.9.10 Hydraulic oil lines. High pressure oil lines shall be steel tubing wherever practical and shall conform to SAE J525. Hydraulic hose assemblies shall also conform to applicable SAE standards and shall meet the age requirements specified in 3.2. Hydraulic hoses shall be provided with reusable hose fittings. "Quick connect/disconnect", pressure-sensing, fittings with protective caps shall be provided on each circuit to facilitate servicing the hydraulic system. These fittings shall be marked for identification of the circuit.

3.3.9.10.1 Hydraulic hoses quick disconnect. Those hoses required for operating the multipurpose bucket shall be equipped with quick disconnect fittings. Dust caps and plugs shall be attached to the fitting at the point of disconnect. The caps and plugs shall be provided for both boom and bucket hoses.

3.3.9.11 Hydraulic cylinders. Hydraulic cylinder rods shall have a hard chromium plating with a crack-free thickness sufficient to withstand a 30-hour saltwater spray test as specified in 4.7.21.

3.3.10 Electrical system. The elctrical power for the excavator shall be supplied by a sufficient number of batteries for the low temperature operation specified herein. The system shall be designed for 24-volt negative ground, and will be resistant to saltwater corrosion.

3.3.10.1 Batteries. Batteries shall be 6TN in accordance with MS35000-3. Batteries shall be furnished charged and dry; electrolyte is not to be provided.

3.3.10.2 Alternator. The alternator shall have a rating of not less than 60 amperes and shall conform to MIL-G-46795.

3.3.10.3 Slave receptacle. The electrical circuit shall incorporate a slave receptacle conforming to MS75058 for starting the engine from auxiliary sources. The receptacle shall be located in a position accessible to a person standing on the ground.

3.3.10.4 Electrical controls. The electrical controls shall include engine shutdown solenoid, horn relay, neutral start switch, outrigger switches and relays and fast hoist solenoids.

3.3.11 Lighting system. The lighting system shall include head lights, tail lights, front and rear turn signal lights, front clearance lights, rear clearance lights, cab and shroud clearance lights, and panel lights in cab and on engine hood panel. Working lights shall also be furnished consisting of one cab, one turntable, and two boom mounted lights.

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3.3.12 Excavating attachments. The following excavating attachments shall be furnished:

BUCKETS	CAPACITY	WIDTH
Excavator	5/8 cubic yard (yd^3)	36 in
Ditch cleanout	7/16 yd^3	60 in
Multipurpose	1-1/2 yd^3	60 in

3.3.12.1 Excavator bucket. The excavator bucket shall be designed for general purpose duty and shall have a minimum struck capacity in accordance with SAE standard J296. The width specified shall be the "cutting width" measured over the bucket side bits. Bucket side cutters shall be provided. Buckets shall be profiled for minimum digging penetration resistance and friction and shall be equipped with hardened wear resistant and replaceable teeth and side cutters on bucket cutting edge at spacing of 6 to 10 inches.

3.3.12.2 Ditch cleanout bucket. The ditch cleanout bucket shall be suitable for cutting "V" ditches, fine grading, sloping and contouring at any angle up to 40 degrees left or right mounted on tool angling device and tool-boom. Minimum requirements are as follows:

Maximum reach at ground level (not less than)	28'0"
Minimum reach at ground level (not more than)	11'0"
Maximum digging depth (not less than)	17'6"
Maximum dump clearance1/ (not less than)	20'9"

The ditch cleanout bucket shall be designed with wide smooth cutting edge without teeth for cutting "V" ditches, contouring and sloping in addition to light duty excavating. The bucket minimum struck capacity shall be in accordance with PCSA standard 3. The mounting shall be identical to other buckets for mounting on the tool angling device or directly on the tool boom extension.

3.3.12.3 Multipurpose bucket A 1-1/2 yd^3 multipurpose front loading 60-inch-wide bucket shall be provided. The capacity shall be SAE struck in accordance with SAE J742B. The bucket shall be profiled for digging penetration and minimum resistance and friction. It shall be equipped with hardened wear resistance cutting edge and replaceable teeth on the bucket edge. Requirements are as follows:

Weight-(lbs) (minimum)	1000
Maximum clam opening width-(in) (not less than)	50

1/Vertical clearance of bucket from grade with tool at maximum height.

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Maximum reach at ground level (not less than)	25'5"
Minimum reach at ground level (not more than)	15'6"
Maximum digging depth (teeth) (not less than)	14'5"
Maximum dump clearance[1/] (not less than)	17'5"
Maximum bucket teeth height[2/] (not less than)	25'5"

3.3.13 Lubricants. All moving parts subject to wear shall be provided with suitable lubrication. The excavator shall be designed to operate efficiently and satisfactorily when lubricated with standard military lubricants conforming to MIL-L-2104 and MIL-L-46167 for engine oils; MIL-G-10924 for automative grease; MIL-L-2105 for gear oil; and MIL-L-2104 for the hydraulic system oil and the transmission/torque converter oil.

3.3.14 Towing attachments. Towing attachments shall be provided on the front and rear of the excavator. The towing attachments shall not reduce the angle of approach and departure of the excavator and shall be capable of withstanding the rigors of retrieving operations from mud or water by a winch and cable in the event of engine failure. Auxiliary airlines and quick-coupling fittings conforming to MS35746-1 shall be installed front and rear to facilitate brake operation during towing.

3.3.14.1 Pintle hook. A rear-mounted pintle hook conforming to MS51118 shall be provided. The pintle hook shall be located on the vertical center line of the excavator not less than 31 nor more than 37 in above ground level. The pintle hook mounting shall be of sufficient strength to withstand, without permanent deformation or damage, the draw-bar pull induced when the pintle hook is restraining forward motion at torque converter stall.

3.3.15 Lifting and tiedown attachments. The excavator shall be equipped with lifting and tiedown attachments conforming to type III of MIL-STD-209. Antirotation steel rods shall be provided for the excavator in order to prevent damages caused by the free rotation of upper structure during shipment. Attachments shall be indicated by a transportation plate conforming to MIL-P-514.

3.3.16 Rifle mounting kit. A rifle mounting kit conforming to Drawing No. 8712407 shall be mounted in a location accessible to the opera tor. The location shall be approved by the contracting officer.

3.3.17 Toolbox. Means shall be provided to carry those tools and accessories required to operate the excavator. The toolbox shall be the manufacturer's standard made of sheet steel not less than 0.050 in

1/ Minimum vertical clearance of bucket teeth from grade with tool at maximum height.

2/ Bucket teeth - distance from grade at end of highest dump.

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nominal thickness (U.S. standard gage number 14). The toolbox shall have a hinged lid and the means to keep the lid closed when subjected to vibration or rough travel. The toolbox shall be fastened to the excavator in a protected and accessible location.

3.3.18 Safety and noise. The excavator shall comply with the latest OSHA requirements applicable to hydraulic wheeled excavators, and shall possess sound suppression as stated in 3.3.6.5. All walkways shall be furnished with a nonslip surface.

3.4 Reliability. Each excavator produced shall be as reliable as the first article.

3.5 Construction.

3.5.1 Steel fabrication. Steel used in the fabrication of the excavator shall be free from kinks and sharp bends. The straightening of material shall be done by methods which will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Reentrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.5.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers or other positive locking shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.5.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.5.4 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack or intermittent welds for strength will not be permitted. Weld penetration shall provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall conform to the procedures as prescribed by either the AWS or the ASME for the type of welding operation to be performed.

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3.5.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings, and otherwise to the standards of good commercial practice.

3.5.6 Castings. Castings shall be sound and free from defects which might render the casting unsound for use.

3.5.7 Threaded parts. Threaded parts shall be in accordance with the applicable SAE fastener standard. All hex bolts and capscrews shall be SAE grade 5 or 8 and shall be so marked.

3.6 Maintainability. The excavator design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field. Access doors shall be capable of being held in the open position. Maintenance shall be accomplished with the use of conventional general purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the excavator.

3.7 Transportability. The excavator shall be designed and constructed so that it can be transported by rail, highway, and water transport. The excavator shall be capable of being loaded and unloaded from an LST (see Figure 1) without disassembly. The excavator components, such as cab, integral counterweight, upper shroud or boom may be removed in order to reduce the excavator height and weight to meet the following maximum limitations:

Height	102 in
Width	96 in
Axle Weight	
front	16,200 lbs
rear	14,900 lbs

3.8 Performance. The excavator shall withstand without damage operation on unimproved terrain; towing by another vehicle; fresh and salt-water fording at water depths to 60 in; and storage as specified herein.

3.8.1 Temperature. Without using auxiliary power from external sources, the excavator shall be equipped to start and operate under continuous digging or road conditions in not more than 30 minutes after engine start in any ambient temperature from -25°F to +125°F. The excavator shall not be damaged by daily storage at the above temperatures or for 30-day periods at temperatures as low as -65°F, or as high as +160°F.

3.8.2 Altitude. The excavators shall be powered with an engine which will provide the power required to perform the requirements specified herein at 5,000-ft altitude.

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LEGEND

BOW RAMP A
CAPACITY-75 TON
LENGTH-100'
WIDTH-15'
MAX RAMP ANGLE-21°
CLEARANCE TO OVERHEAD-NOT APPLICABLE

BETWEEN DECK RAMP B
CAPACITY-75 TON
CLEARANCE TO OVERHEAD:
RAMP DOWN-13'3"
RAMP UP-13'7"
WIDTH-15'8"
LENGTH-47'9"
RAMP ANGLE-19°

DECK RAMP
BOTTOM-20'
TOP-10'
HEIGHT-1'

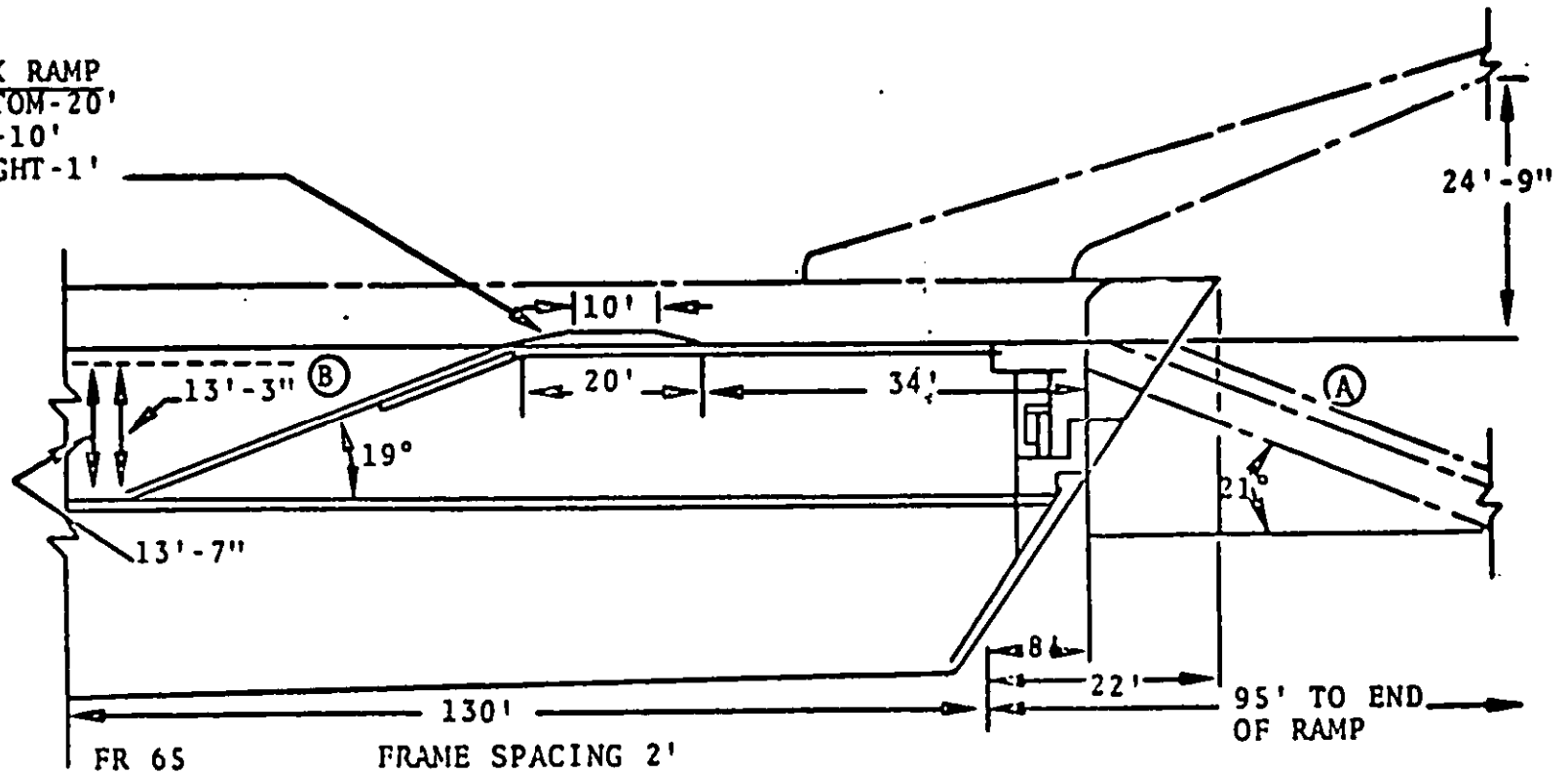


Figure 1. Profile, LST 1179 class bow

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3.8.3 Rain and wind. Excavators shall be designed to withstand extended periods of high winds and heavy rains during operation and storage. Under these conditions, there shall be no evidence of leakage into the intake filter, exhaust system, engine crankcase, hydraulic or other reservoirs and the engine shall start in 2 minutes after such exposure. Excavator shall be tested in accordance with 4.7.18.

3.8.4 Water fording capability. The excavator shall have the inherent capability of fording fresh or salt water to a depth of 60 in without damage. When tested after standing 1 hour in 60 in of water, the amount of water in the engine, clutch, transmission, torque converter or axle housings shall not be more than 1 percent of the total oil content, nor will any systems be adversely affected. All components, assemblies, and systems of the excavator which would be damaged or could cause malfunction of the excavator when exposed to salt water shall be enclosed in a water resistant housing or jacket. Equipment enclosed in a water resistant housing which must be are-vented shall have a vent tube or pipe extending above the 60-in fording depth to preclude the entry of water. The excavator shall be capable of being started at full water depth. All exposed components subject to contamination or corrosion from fresh or salt water shall be protected by use of corrosion-resistant plating or corrosion-resistant treatment, except the wearing surfaces of functional chains, rollers, and slides, which shall be capable of being lubricated. The application of corrosion-resistant plating or treatment shall give protection to any part for its functional life. There shall be no evidence of water damage to, or water leaking into, water-resistant housings, water-resistant jackets, and closed tanks of the engine transmission, torque converter, fuel system, brakes, and hydraulic system or any evidence of water damage or corrosion of any excavator part which affects proper operation of the excavator and its ability to meet any of the requirements specified herein.

3.8.5 Travel gradeability. The excavator shall have the ability to travel at a stabilized speed of 2 mi/h on smooth, firm, dry ground grades with tool unloaded and boom in travel position and to stop on the grades using the service brake. The excavator shall ascend and descend 35 percent grades, and travel to the right and left on a grade of 30 percent.

3.9 Electromagnetic interference. The electromagnetic radiation limits of the excavator shall conform to MIL-STD-461, Class IIIC.

3.10 Dimensions. Dimensions shall be as specified in 3.3.1.

3.11 Weight. Weight shall be as specified in 3.3.1.

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3.12 Color and finish. The surfaces normally painted in good commercial practice shall be cleaned, treated and painted as specified in MIL-T-704, type A. The finished coat shall be in accordance with MIL-E-52798 (forest green).

3.13 Identification markings. Each individual excavator and each separate attachment removable for either surface or air transportability or other purpose shall be identified in accordance with MIL-STD-130. A U.S. Marine Corps nameplate shall be permanently attached in a conspicuous location. The nameplates will be furnished upon request, by the Commandant of the Marine Corps (Code LMA-3), Headquarters U.S. Marine Corps, Washington, D.C. 20380. Data applicable to blank spaces 1, 3, 11, and 12 will be furnished by the procuring activity. The contractor shall be required to imprint the data in all blank spaces. All other plates other than the name plate, shall be provided by the manufacturer and shall confrom to MIL-P-514, type 1, grade A, class 1. Each plate shall be attached by screws, drive screws, bolts or rivets in a conspicuous, protected location.

3.13.1 Instruction plates. Each excavator shall be equipped with instruction plates, including warnings and cautions, suitably located describing any special or important procedures to be followed in operating, servicing, or dissassembling the excavator. Plates shall be of a material which will last and remain legible for the life of the excavator, and shall be securely affixed thereto with brass screws of not less than 1/8-inch diameter or by other means.

3.13.2 Registration data. The registration number assigned to each excavator by the contracting officer shall be affixed on the excavator by the contractor using adhesive backed decals, nonreflective, conforming to type II, class I of MIL-M-43719. Registration numbers shall be applied on right and left sides, as well as the front and rear of the machine in minimum 3-inch characters. Color shall be black.

3.13.3 Stenciling. The gross weight of each excavator shall be stenciled on each side of the excavator in such a manner as to be readily discernible to dock and military personnel. The prescribed tire pressure shall be stenciled on the carrier fenders by block or stencil-type letters not more than 1-inch high. Decals conforming to type II, class I of MIL-M-43719 shall be utilized. Color shall be black.

3.14 Workmanship. All parts, components, assemblies of the excavator, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free of burrs, sharp edges, and corners except when sharp edges or corners are required.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3)
(b) Production (see 4.4)
(c) Preparation for delivery (see 4.5)

4.3 First article. First article inspection shall be performed on the first production item. This inspection shall include the examination in 4.6 and the tests in 4.7. The first article may be a standard production item from the supplier's current inventory, provided the excavator meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining excavators to be furnished under the contract.

4.4 Production. Each excavator produced in fulfillment of a Government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warm-up period as recommended by the manufacturer. Each excavator shall be subjected to examination in 4.6. The excavator shall be given a run-in test and all controls shall be operated a sufficient number of times to assure that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction.

4.5 Packaging. The preservation, packaging, packing, and marking of the excavator shall be inspected to verify conformance to the requirements of Section 5.

4.6 Examination. Each excavator shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with requirements specified herein or any necessary redesign or modification following failure to meet requirements specified herein shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any requirements specified herein or presence of one or more defects preventing or lessening efficiency shall constitute cause for rejection.

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4.7 Tests. Failure of the excavator to pass any of the following tests or failure of any component during testing shall be the basis for rejection.

4.7.1 Control function. All operating controls and safety interlocks specified shall be identified and operated. Inability to meet the requirements specified herein shall constitute failure of this test.

4.7.2 Characteristics. The characteristics specified in 3.3.1, 3.3.2, and 3.3.4 shall be verified. Certified test reports of tests accomplished in accordance with applicable SAE procedures will be accepted as they apply to the unit procured.

4.7.3 Gradeability. The excavator shall be given the tests necessary to establish conformance with 3.8.5. Inability to travel on the specified grades or for the service brakes to hold on the specified grades shall constitute failure of this test.

4.7.4 Engine horsepower. The manufacturer shall certify that the engine manufacturer has approved the engine for application as specified herein and shall furnish (1) a certified copy of the dynamometer test performed on a sample engine per SAE J816, (2) a performance curve of fully equipped engine as installed in the in the excavator, and (3) derating factors applicable when the engine is operated at altitudes above those specified in 3.8.2, and ambient temperatures above 85°F.

4.7.5 Drawbar pull test. Drawbar pull tests shall be conducted to determine conformance to 3.3.2. Drawbar pull less than specified in 3.3.2 shall constitute failure of this test.

4.7.6 Road test. The excavator shall be given a road test, not exceeding engine governed speed, for a distance of not less than 25 miles as follows:

(a) Travel 15 miles over paved roads without load at average speeds of not less than 20 mi/h.

(b) Travel 10 miles over uneven roads with rough terrain containing 30 percent up slopes at an average speed of not less than 3 mi/h without load.

Inability to complete the road test at the specified average speeds shall constitute failure of this test.

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4.7.7 Service brake test. Service brakes shall be tested for compliance with 3.3.7.2 and in accordance with SAE J1152 for excavators in this class. This test shall be performed with both wet and dry brakes (see 4.7.13.e). Immediately after the excavator comes to a halt, this test cycle shall be repeated for four additional cycles. The above test cycle shall be repeated five additional times traveling in the opposite direction of the start of the test. Where there is indication of damage, all shoes, drums, and other brake components shall be disassembled and examined. Evidence of damage which could result in unsafe operating conditions shall constitute failure of this test.

4.7.8 Parking brake test. The parking brake shall be tested for compliance with 3.3.7.3. Inability to hold on a 30 percent slope shall constitute failure failure of this test.

4.7.9 Hydraulic system test. The engine and all operating components shall be given a warmup operation for a continuous period of not less than 30 minutes, during which all of the functions enumerated in 3.3.9 shall be checked to ascertain the functioning of the hydraulic system. Incorrect functioning of any component shall constitute failure of this test.

4.7.10 Hydraulic hose test. Hydraulic hoses and fittings shall be bench tested to determine conformance to 3.3.9.10, except that certification by the manufacturer, or certified copies of test reports on identical items previously tested for identical pressures may be accepted.

4.7.11 Operational tests with accessories. The excavator, together with 7/16-yd^3 excavating bucket, located in rough terrain shall be tested under full load conditions to ascertain compliance with the requirements of 3.3.2, 3.3.4, and 3.3.12. During this test, observance shall be made to ascertain any instability, improper functions, leakages, and overloading in the hydraulic system of the excavator. Each optional bucket, furnished under the contract, shall be assembled on the boom and operated throughout its functional range under full load simulating intended service conditions. Any failure of components or distortion of parts shall be cause for rejection. The first accessory shall be tested for 8 hours; all other accessories shall be tested for 4 hours each.

4.7.12 Lifting capacity tests. Lift tests shall be conducted on the excavator in accordance with SAE J1097. The balance point capacity in the least stable direction shall be determined at each radius specified in 3.3.3.

4.7.13 Travel control and turning circle test. To demonstrate workability of the travel controls (3.3.6.3), the excavator shall be operated forward and backward in two figure 8 patterns with maximum right and maximum left turns. The excavator shall then negotiate a turning circle in compliance with SAE J695. Inability to negotiate figure 8 patterns or a vehicle clearance circle in excess of 80 ft shall constitute failure of this test.

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4.7.14 Water fording capability tests. Water fording capabilities in compliance with 3.8.4 shall be ascertained in accordance with the following and Table II. A fresh or saltwater source may be used for the test.

(a) Bring excavator to operating temperature, and sample oil from the engine and all drive compartments for control purposes.

(b) Operate a minimum of 2 hours in 60 in of water; immediately thereafter, operate onshore in all gears for 2 hours. The service brake test in 4.7.7 shall be run at the beginning of each 2-hour land test.

(c) Repeat a and b above four additional times for a total of 20 hours, except as follows:

(1) On succeeding tests, the excavator engine shall be started and stopped a minimum of four times in 60-in deep water.

(2) Final (fifth) fording shall consist of 3 hours in the water and 1 hour on shore.

(3) Oil samples shall be taken during the first, second, and fifth tests for analysis of the allowable water content as specified in 3.8.4.

(d) Check running gear (wheel brakes) for compliance with 4.7.7 and for any deficiencies which could cause unsafe operation. Inability to start the engine in the water, water in excess of allowable content in the oil or inability of the excavator to complete the test shall constitute failure of this test.

(e) Note and report all deficiencies.

Table II - Water Fording Capabilities

Test	Water	Shore	Oil Sample	Start/Stop
1	2 hr	2 hr	Yes	
2	2 hr	2 hr	Yes	4 times
3	2 hr	2 hr	No	4 times
4	2 hr	2 hr	No	4 times
5	3 hr	1 hr	Yes	4 times

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4.7.14 Towing attachment(s) test. The excavator shall be placed in water at a depth of 60 in with the engine shut off. A cable shall be attached to the towing attachment(s), and the excavator shall be pulled from the water without use of its own power. This test shall be performed on both front and rear towing attachments. Breakage, distortion or deformation of the towing attachments or excavator shall constitute failure of this test.

4.7.16 Lifting and tiedown attachments test. The excavator shall be tested to verify that the attachments conform to the requirements specified in 3.3.15. Failure of any component, or breakage, distortion or deformation shall constitute of this test.

4.7.17 Test for electromagnetic interference characteristics. The excavator shall be tested in accordance with MIL-STD-462 for Class III equipment. Inability to pass the test shall constitute failure of this test.

4.7.18 Wind and rain test. The wind and rain test shall be conducted as follows:

(a) TEST APPARATUS:

(1) Timing device.

(2) Measuring device (U.S. Weather Bureau type preferred).

(3) Spray fixture.

(b) TEST PROCEDURE:

(1) Subject the excavator to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the excavator at an angle of 45 degrees, ±5 degrees, with a force equivalent to a 30 mi/h wind. The simulated rainfall shall disperse uniformly over the surfaces of the excavator.

(2) Expose the top and left side of the excavator to the rain described above for a period of 5 minutes with engine off. With rain still flowing, start engine and allow engine to idle for 2 minutes.

(3) Repeat (b) with the top and front exposed.

(4) Repeat (b) with top and right exposed.

(5) Repeat (b) with top and rear exposed.

(6) Stop the rainfall and allow the excavator to stand for not less than 30 minutes with the engine off. Drain a small quantity of fluid from each fluid reservoir (except cooling system) and examine for presence of water.

(7) Check electrical components, gages, cab (if furnished), and the air cleaner for water entry and damage.

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4.7.19 Pintle hook test. The pintle hook shall be tested by exerting the maximum drawbar pull on it for a period of not less than 5 minutes. Any distortion or breakage shall constitute failure of this test.

4.7.20 Sound level test. The excavator shall be tested to determine conformance with 3.3.6.5 and SAE J1166. Inability to meet these requirements shall constitute failure of this test.

4.7.21 Salt spray test. Certification is acceptable for the requirements as identified in 3.3.9.11.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The complete excavator shall be preserved and packaged in accordance with level A of MIL-C-3580.

5.1.2 Level C. The complete excavator shall be preserved and packaged in accordance with the contractor's standard practice.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The complete excavator shall be packed in accordance with level A of MIL-C-3580, except that the telescoping boom shall be secured in the boom rack in the normal travel position.

5.2.2 Level B. The complete excavator shall be packed in accordance with level B of MIL-C-3580.

5.2.3 Level C. The complete excavator shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification Rules, or National Motor Freight Classification Rules.

5.3 Marking. The excavators, attachments, interior packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The excavator described herein shall be used in military operations involving digging undisturbed soils and rock and handling loose materials, as well as loading/unloading fortification materials, placing bunkers and preformed shelters into excavated spaces, and other uses which are within its capabilities. The excavator shall be subject to the riggers of amphibious loading and unloading from naval vessels, as well as sea transport and other conditions inherent to military operations.

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6.2 Ordering data. Procurement data should specify:

6.2.1 Procurement requirements:

(a) Title, number and date of this specification.

(b) Level of preservation, packaging and packing (5.1 and 5.2)

6.6.6 Contract data requirements. Data conforming to Data Item Description DI-E-4527, DI-M-4704 and DI-R-4802 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 First article. Any changes or deviations of production excavators from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of his obligation to furnish excavators conforming to this specification.

Preparing activity:
Navy - MC
Project No. 3805-N099

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-LF-014-1802

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